

Kaizar Rangwala

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EDUCATION:

University of California, Berkeley

Major - Computer Science, BA;

Minors - Bioengineering, Global Public Health

Berkeley, CA

Expected Graduation: May 2029

Relevant Coursework:

CS Technical - Discrete Math & Probability, Data Structures, Program Structures, Linear Algebra and DiffEq

Premedical - Organic Chemistry, General Chemistry, General Biology, Biostatistics, Epidemiology

Honors, Awards, Recognitions:

- 1st person in the United States under age 20 to publish SciML research
- Samsung SFT National Delegate — \$12,000 Prize for STEM Research Funding
- National Merit Scholar — \$2,500 Scholarship for Undergraduate Education

SKILLS:

Programming Languages: Python, Java, SQL, Julia, Scheme

Machine Learning: Pandas, NumPy, OpenCV, Scikit, PyTorch, PyRadiomics, Tensorflow, Matplotlib, Seaborn

Software Engineering: OOP, AWS, Git, Flask, Selenium, REST APIs, Bootstrap

Clinical Certifications: EMT-B, Bloodborne Pathogen Management, Nebulizer, Naloxone, CPR, Glucometer

EXPERIENCE:

Clinical Data Animation Center, Stanford Neurology: ML Researcher

- Developed LLM-guided clinical decision support system for epilepsy surgical prognosis
- Integrated large language model outputs with patient data to inform neurosurgical planning
- Built ML pipeline to analyze seizure and imaging data for pre-operative outcome prediction
- Collaborated with Stanford neurologists to validate model predictions against clinical benchmarks

Visa: Computational Finance Analyst (Contract)

February 2026 - May 2026

- Built a Python AI agent framework to analyze transactional data and benchmark institutional performance
- Integrated OpenAI APIs into structured pipelines to evaluate and score bank authorization processes
- Automated drafting of financial proposals and contract recommendations from quantitative benchmarking outputs
- Researched bank authorization workflows to identify inefficiencies and refine agent decision logic

Amazon Web Services: Machine Learning Engineer (Contract)

September 2025 - December 2025

- Automated a pipeline to collect novel ML Research, implement it in PyTorch, and test fidelity on AWS Trainium
- Built Scraper for ML research with AWS Lambda, SQS, Opensearch, and built a Claude RAG model for filtering
- Utilized Claude via Bedrock to generate Pytorch implementations of selected novel ML techniques
- Designed a Flask App to execute code on TRN and evaluate performance with Neuron Profiler & Perfetto

Duke University Neurosurgical Unit: Translational ML Researcher

April 2024 - May 2025

- Studied an association between a tumor's 3-dimensional shape and underlying cancerous genetic mutations
- Used radiomics to extract and quantify geometric features of glioma and meningioma MRIs into a dataset
- Trained an XGBoost model to predict presence of mutations using feature dataset, achieved AUROC of 0.949
- Demonstrated potential of ML & radiomics to guide surgical strategy, surveillance, and patient counseling

SELECTED PUBLICATIONS & CONFERENCE PROCEEDINGS:

Enhanced Modeling of Harmful Algal Bloom (HAB) Growth Using Scientific Machine Learning

Presented at MIT Undergraduate Research Technology Conference, Published in IEEE Xplore

ML with Radiomics To Predict WHO Pathologic Grade of Meningioma from Preoperative MRI

Published in Journal of Neurosurgery, Presented at North American Skull Base Society Conference (Podium)